

PHYSICS PRACTICAL SHEETS

KU Campus -

Date: 14 May, 2026

Class: CE-I/II

Roll No.: 43

Shift: Morning

Object of the Experiment (Block Letter)

Experiment No.: 7

Group: S

Sub: PHYS 102 Lab

Set: I

Double check

STUDY OF THE PHENOMENON OF RESONANCE IN A SERIES LCR CIRCUIT
AND DETERMINE THE RESONANCE FREQUENCY, AND QUALITY FACTOR

Apparatus required:

- 1) LCR series & Parallel Resonance Circuit kit
- 2) AC power supply
- 3) connecting wires

Theory:

In a series LCR circuit, resonance occurs when the inductive reactance (X_L) equals the capacitive reactance (X_C):

$$X_L = X_C \quad \text{--- (1)}$$

At resonance: The impedance of the circuit is minimum, resulting in maximum current flow. The current is maximum at the resonant frequency (f_0) & decreases as the frequency moves away from f_0 . This variation is characterized by the sharpness of the resonance curve, which is determined by the quality factor (Q).

Resonant frequency:

The resonant frequency (f_0) is given by:

$$f_0 = \frac{1}{2\pi\sqrt{LC}} \quad \text{--- (2)}$$

where, L is the inductance of the inductor, & C is the capacitance of the capacitor.

Quality factor (Q)

The quality factor (Q) measures the sharpness of the resonance & is expressed as:

$$Q = \frac{1}{R} \sqrt{\frac{L}{C}} \quad \text{--- (3)}$$

where, R is the resistance of the circuit.

Table 1: For recording frequency and current at varying resistance in the study of resonance in a series LCR circuit.

No of obs.	$R_1 = 50 \Omega$		$R_2 = 100 \Omega$		$R_3 = 150 \Omega$	
	Frequency (Hz)	Current (I)	Frequency (Hz)	Current (I)	Frequency (Hz)	Current
1	1	4	1	4	1	4
2	1.5	5	1.5	4.5	1.5	4.5
3	2	9	2	8.5	2	7.5
4	2.5	14	2.5	11.5	2.5	9.5
5	3	17.5	3	13.5	3	11
6	3.5	18.5	3.5	14	3.5	11.5
7	4	16.5	4	13.5	4	10.5
8	4.5	14.5	4.5	12	4.5	10
9	5	13.5	5	11	5	9.5
10	5.5	12	5.5	10	5.5	9
11	6	10.5	6	9.5	6	8.5
12	6.5	9.5	6.5	9	6.5	8
13	7	9	7	8	7	7.5
14	7.5	8.5	7.5	7.5	7.5	7
15	8	8	8	7	8	6.5
16	8.5	7.5	8.5	6.5	8.5	6

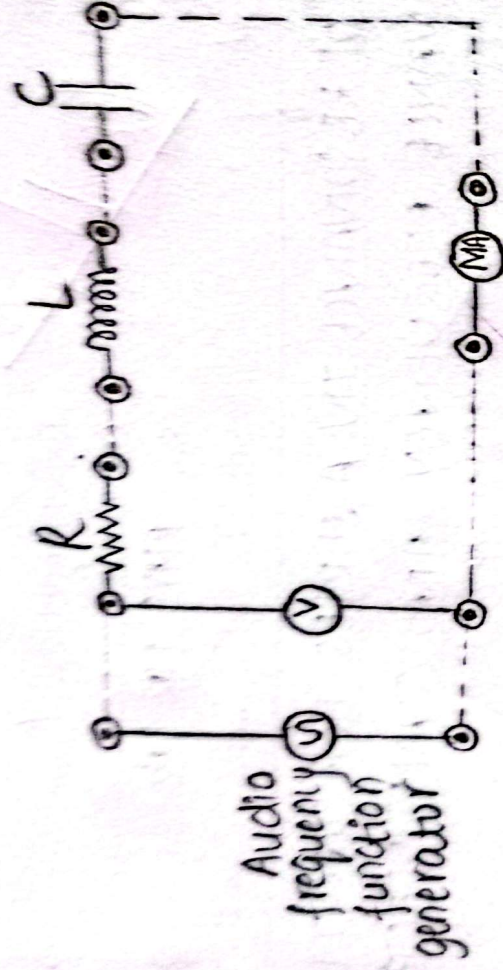


Fig 1: LCR series Resonance circuit

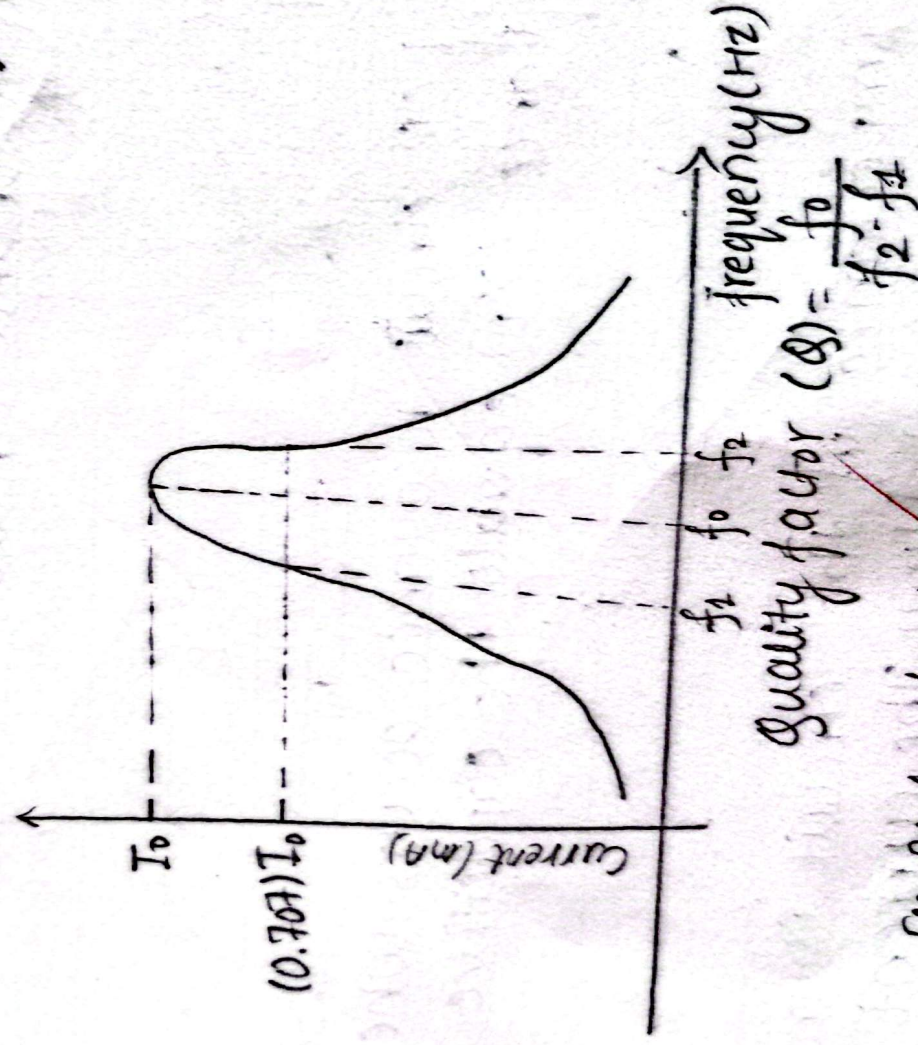


Fig 2: Graph of current vs frequency

Table 2: For the calculation of Resonant frequency and the quality factor.

No. of obs	Resistance	Resonant frequency (Hz)		Quality factor (Q)	
		$f_0 = \frac{1}{2\pi\sqrt{LC}}$	from graph	$Q = \frac{1}{R}\sqrt{\frac{L}{C}}$	from graph
1	50	3.4	3.5	4.26	1.75
2	100	3.4	3.5	2.13	1.2
3	150	3.4	3.5	1.42	0.81

Result: The resonant frequency is 3.4 Hz and quality factor is 1.25.

~~Precautions:~~

Precautions:

1) Test the continuity and functionality of wires and components using a Multimeter.

2) Turn off the the power supply before making or modifying any circuit connections.

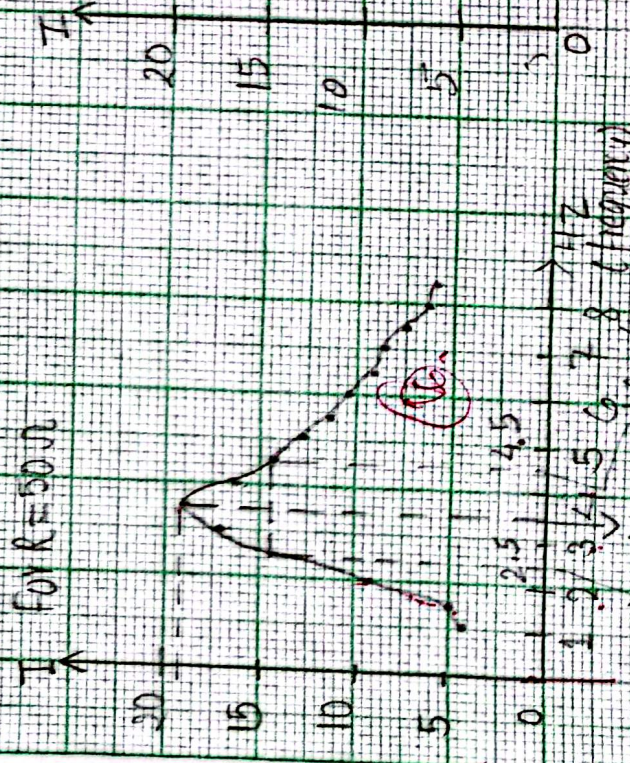
3) Handle capacitors and inductors carefully to avoid damage.

4) Avoid touching live terminals during the experiment to prevent electric shock.

Conclusion: Hence the resonance frequency & quality factor were found.

Scaling: In x axis: 10 small division = 2 Hz
In y axis: 10 small division = 5 A

FOR $R = 50 \Omega$



$$\text{Quality factor} = \frac{f_0}{f_2 - f_1} = \frac{7.5}{4.5 - 3.5} = 7.5$$

FOR $R = 150 \Omega$



$$\text{Quality factor} = \frac{f_0}{f_2 - f_1} = \frac{7.5}{6.5 - 5.5} = 7.5$$

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Fig 3: Graph of current vs frequency